VTU Internship Report Document

This PDF contains the uploaded internship report code/document content.

```
const {
  Document, Packer, Paragraph, TextRun, Table, TableRow, TableCell,
  Header, Footer, AlignmentType, HeadingLevel, BorderStyle, WidthType,
  ShadingType, VerticalAlign, PageNumber, PageBreak, LevelFormat,
  TableOfContents, UnderlineType
} = require('docx');
const fs = require('fs');

// ■■■ Border helpers ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
const PAGE_BORDER = { style: BorderStyle.SINGLE, size: 24, color: "000000", space: 20 };
const NO_BORDER   = { style: BorderStyle.NONE,   size: 0,  color: "FFFFFF" };
const THIN_BORDER = { style: BorderStyle.SINGLE, size: 4,  color: "CCCCCC" };
const THIN_BORDERS = { top: THIN_BORDER, bottom: THIN_BORDER, left: THIN_BORDER, right: THIN_BORDER };

// ■■■ Helpers ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
const center = (text, opts = {}) => new Paragraph({
  alignment: AlignmentType.CENTER,
  spacing: { before: opts.before ?? 80, after: opts.after ?? 80 },
  children: [new TextRun({ text, font: "Times New Roman", ...opts })]
});

const centeredBold = (text, size = 24, opts = {}) =>
  center(text, { bold: true, size, ...opts });

const body = (text, opts = {}) => new Paragraph({
  alignment: opts.align ?? AlignmentType.JUSTIFIED,
  spacing: { before: opts.before ?? 60, after: opts.after ?? 60, line: 360, lineRule: "auto" },
  indent: opts.indent ? { firstLine: 720 } : {},
  children: [new TextRun({ text, font: "Times New Roman", size: 22, ...opts })]
});

const heading1 = (text) => new Paragraph({
  heading: HeadingLevel.HEADING_1,
  alignment: AlignmentType.CENTER,
  spacing: { before: 200, after: 200 },
  border: { bottom: { style: BorderStyle.SINGLE, size: 6, color: "000000", space: 4 } },
  children: [new TextRun({ text: text.toUpperCase(), font: "Times New Roman", bold: true, size: 26 })]
});

const heading2 = (text) => new Paragraph({
  heading: HeadingLevel.HEADING_2,
  spacing: { before: 180, after: 120 },
  children: [new TextRun({ text, font: "Times New Roman", bold: true, size: 24 })]
});

const heading3 = (text) => new Paragraph({
  heading: HeadingLevel.HEADING_3,
  spacing: { before: 120, after: 80 },
  children: [new TextRun({ text, font: "Times New Roman", bold: true, underline: { type: UnderlineType.SINGLE }
});

const bullet = (text) => new Paragraph({
  numbering: { reference: "bullets", level: 0 },
  spacing: { before: 40, after: 40, line: 320, lineRule: "auto" },
  children: [new TextRun({ text, font: "Times New Roman", size: 22 })]
});

const numbered = (text) => new Paragraph({
  numbering: { reference: "numbers", level: 0 },
  spacing: { before: 40, after: 40, line: 320, lineRule: "auto" },
  children: [new TextRun({ text, font: "Times New Roman", size: 22 })]
});

const subbullet = (text) => new Paragraph({
  numbering: { reference: "subbullets", level: 0 },
  spacing: { before: 30, after: 30, line: 300, lineRule: "auto" },
  children: [new TextRun({ text, font: "Times New Roman", size: 22 })]
});

const sp = (n = 1) => [...Array(n)].map(() => new Paragraph({
  spacing: { before: 0, after: 0 },
```

```
  children: [new TextRun({ text: "" })]
}));

const pageBreak = () => new Paragraph({
  children: [new PageBreak()]
});

const divider = () => new Paragraph({
  spacing: { before: 80, after: 80 },
  border: { bottom: { style: BorderStyle.SINGLE, size: 4, color: "000000", space: 1 } },
  children: [new TextRun({ text: "" })]
});

// ■■■ TOC Table ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function tocRow(chapter, title, page) {
  const nb = { style: BorderStyle.NONE, size: 0, color: "FFFFFF" };
  const brd = { top: nb, bottom: nb, left: nb, right: nb };
  return new TableRow({
    children: [
      new TableCell({ borders: brd, width: { size: 1000, type: WidthType.DXA }, margins: { top: 40, bottom: 40,
        children: [new Paragraph({ children: [new TextRun({ text: chapter, font: "Times New Roman", size: 22 })
      new TableCell({ borders: brd, width: {

size: 6960, type: WidthType.DXA }, margins: { top: 40, bottom: 40, left: 80, right: 80 },
        children: [new Paragraph({ children: [new TextRun({ text: title, font: "Times New Roman", size: 22 })]
      new TableCell({ borders: brd, width: { size: 1400, type: WidthType.DXA }, margins: { top: 40, bottom: 40,
        children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: page, font:
    ]
  });
}

function figRow(fig, title, page) {
  const nb = { style: BorderStyle.NONE, size: 0, color: "FFFFFF" };
  const brd = { top: nb, bottom: nb, left: nb, right: nb };
  return new TableRow({
    children: [
      new TableCell({ borders: brd, width: { size: 1200, type: WidthType.DXA }, margins: { top: 40, bottom: 40,
        children: [new Paragraph({ children: [new TextRun({ text: fig, font: "Times New Roman", size: 22 })] })
      new TableCell({ borders: brd, width: { size: 6760, type: WidthType.DXA }, margins: { top: 40, bottom: 40,
        children: [new Paragraph({ children: [new TextRun({ text: title, font: "Times New Roman", size: 22 })]
      new TableCell({ borders: brd, width: { size: 1400, type: WidthType.DXA }, margins: { top: 40, bottom: 40,
        children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: page, font:
    ]
  });
}

// ■■■ Section builder ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
function section(kids, pageBorder = true) {
  const base = {
    properties: {
      page: {
        size: { width: 12240, height: 15840 },
        margin: { top: 1440, right: 1080, bottom: 1440, left: 1080 },
        borders: pageBorder ? {
          pageBorderTop:    PAGE_BORDER,
          pageBorderBottom: PAGE_BORDER,
          pageBorderLeft:   PAGE_BORDER,
          pageBorderRight:  PAGE_BORDER,
        } : {}
      }
    },
    children: kids
  };
  return base;
}

// ■■■ DOCUMENT ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
const doc = new Document({
  numbering: {
    config: [
      { reference: "bullets",
        levels: [{ level: 0, format: LevelFormat.BULLET, text: "•", alignment: AlignmentType.LEFT,
          style: { paragraph: { indent: { left: 720, hanging: 360 } } } }] },
      { reference: "subbullets",
        levels: [{ level: 0, format: LevelFormat.BULLET, text: "➢", alignment: AlignmentType.LEFT,
          style: { paragraph: { indent: { left: 900, hanging: 360 } } } }] },
      { reference: "numbers",
        levels: [{ level: 0, format: LevelFormat.DECIMAL, text: "%1.", alignment: AlignmentType.LEFT,
          style: { paragraph: { indent: { left: 720, hanging: 360 } } } }] },
```

```
      ]
    },
    styles: {
      default: { document: { run: { font: "Times New Roman", size: 22 } } },
      paragraphStyles: [
        { id: "Heading1", name: "Heading 1", basedOn: "Normal", next: "Normal", quickFormat: true,
          run: { size: 26, bold: true, font: "Times New Roman" },
          paragraph: { spacing: { before: 200, after: 200 }, outlineLevel: 0, alignment: AlignmentType.CENTER } ]
        { id: "Heading2", name: "Heading 2", basedOn: "Normal", next: "Normal", quickFormat: true,
          run: { size: 24, bold: true, font: "Times New Roman" },
          paragraph: { spacing: { before: 160, after: 100 }, outlineLevel: 1 } },
        { id: "Heading3", name: "Heading 3", basedOn: "Normal", next: "Normal", quickFormat: true,
          run: { size: 22, bold: true, font: "Times New Roman" },
          paragraph: { spacing: { before: 120, after: 80 }, outlineLevel: 2 } },
      ]
    },
    sections: [
      // ████████████████████████████████████████████████████████████████████
      // PAGE 1 – TITLE PAGE
      // ████████████████████████████████████████████████████████████████████
      section([
        ...sp(1),
        centeredBold("VISVESVARAYA TECHNOLOGICAL UNIVERSITY", 28),
        cen

teredBold("BELAGAVI-590 018, KARNATAKA", 26),
        ...sp(2),
        centeredBold("AN INTERNSHIP REPORT", 26, { before: 100, after: 100 }),
        centeredBold("ON", 24),
        ...sp(1),
        centeredBold("DATA SCIENCE AND MACHINE LEARNING", 28, { underline: { type: UnderlineType.SINGLE }, before
        ...sp(1),
        new Paragraph({
          alignment: AlignmentType.CENTER,
          spacing: { before: 60, after: 60 },
          children: [new TextRun({ text: "Submitted in the partial fulfillment of the requirements for the award
        }),
        ...sp(1),
        centeredBold("BACHELOR OF ENGINEERING", 24),
        centeredBold("IN", 22),
        centeredBold("COMPUTER SCIENCE AND ENGINEERING", 24),
        ...sp(2),
        new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 60, after: 20 },
          children: [new TextRun({ text: "Submitted  by", font: "Times New Roman", size: 22, italics: true })] })
        centeredBold("SACHIN", 26),
        centeredBold("1BC22CS042", 24),
        ...sp(1),
        new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 60, after: 20 },
          children: [new TextRun({ text: "Under the Guidance of", font: "Times New Roman", size: 22, italics: tru
        centeredBold("Dr. RAMESH KUMAR B", 24),
        center("Assistant Professor", { size: 22, before: 40, after: 20 }),
        centeredBold("Dept. of CSE", 22),
        ...sp(2),
        centeredBold("BANGALORE COLLEGE OF ENGINEERING AND TECHNOLOGY", 24),
        centeredBold("AFFILIATED TO VTU, BELAGAVI", 24),
        ...sp(1),
        centeredBold("DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING", 24),
      ]),

      // ████████████████████████████████████████████████████████████████████
      // PAGE 2 – CERTIFICATE
      // ████████████████████████████████████████████████████████████████████
      section([
        centeredBold("BANGALORE COLLEGE OF ENGINEERING AND TECHNOLOGY", 24),
        centeredBold("AFFILIATED TO VTU, BELAGAVI", 24),
        ...sp(1),
        centeredBold("DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING", 24),
        ...sp(2),
        new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 120, after: 120 },
          border: { bottom: { style: BorderStyle.SINGLE, size: 4, color: "000000", space: 1 } },
          children: [new TextRun({ text: "CERTIFICATE", font: "Times New Roman", bold: true, size: 32, underline:
        ...sp(1),
        new Paragraph({
          alignment: AlignmentType.JUSTIFIED,
          spacing: { before: 80, after: 80, line: 360, lineRule: "auto" },
          children: [
            new TextRun({ text: "Certified that the work entitled ", font: "Times New Roman", size: 22 }),
            new TextRun({ text: '"DATA SCIENCE AND MACHINE LEARNING"', font: "Times New Roman", size: 22, bold: t
            new TextRun({ text: " carried out by ", font: "Times New Roman", size: 22 }),
```

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      new TextRun({ text: "SACHIN (1BC22CS042)", font: "Times New Roman", size: 22, bold: true }),
      new TextRun({ text: " Bonafide student of ", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Bangalore College Of Engineering & Technology, Bangalore", font: "Times New Roma
      new TextRun({ text: " in partial fulfillment for the award of ", font: "Times New Roman", size: 22 })
      new TextRun({ text: "Bachelor of Engineering", font: "Times New Roman", size: 22, bold: true }),
      new TextRun({ text: " in ", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Computer Science and Engineering", font: "Times New Roman", size: 22, bold: true
      new TextRun({ text: " of the ", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Visvesvaraya Technological University, Belagavi", font: "Times New Roman", size:
      new TextRun({ text: " during the academic year ", font: "Times New Roman", size: 22 }),

      new TextRun({ text: "(2025-26)", font: "Times New Roman", size: 22, bold: true }),
      new TextRun({ text: ". It is certified that all corrections/suggestions indicated for the Internship
    ]
  }),
  ...sp(3),
  // Signature row using tab stops
  new Paragraph({
    spacing: { before: 80, after: 20 },
    children: [
      new TextRun({ text: "Signature of Coordinator", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Signature of HOD", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Signature of Principal", font: "Times New Roman", size: 22 }),
    ]
  }),
  new Paragraph({
    spacing: { before: 20, after: 20 },
    children: [
      new TextRun({ text: "Dr. Ramesh Kumar B", font: "Times New Roman", size: 22, bold: true }),
      new TextRun({ text: "\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Mrs. Kavitha R", font: "Times New Roman", size: 22, bold: true }),
      new TextRun({ text: "\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Dr. Gopalaiah", font: "Times New Roman", size: 22, bold: true }),
    ]
  }),
  new Paragraph({
    spacing: { before: 0, after: 20 },
    children: [
      new TextRun({ text: "BE, ME(Ph.D)", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "BE, ME(Ph.D)", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "M.Tech, Ph.D, MISTE", font: "Times New Roman", size: 22 }),
    ]
  }),
  new Paragraph({
    spacing: { before: 0, after: 60 },
    children: [
      new TextRun({ text: "Dept. of CSE", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "Dept. of CSE", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "\t\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "BCET", font: "Times New Roman", size: 22 }),
    ]
  }),
  ...sp(2),
  new Paragraph({ spacing: { before: 80, after: 20 },
    children: [new TextRun({ text: "EXTERNAL VIVA:", font: "Times New Roman", bold: true, size: 22 })] }),
  ...sp(1),
  new Paragraph({ spacing: { before: 20, after: 20 },
    children: [
      new TextRun({ text: "NAME OF THE EXAMINERS", font: "Times New Roman", bold: true, size: 22 }),
      new TextRun({ text: "\t\t\t\t", font: "Times New Roman", size: 22 }),
      new TextRun({ text: "SIGNATURE OF THE EXAMINERS", font: "Times New Roman", bold: true, size: 22 }),
    ]
  }),
  ...sp(1),
  new Paragraph({ spacing: { before: 20, after: 40 }, children: [new TextRun({ text: "1.", font: "Times New
  new Paragraph({ spacing: { before: 20, after: 40 }, children: [new TextRun({ text: "2.", font: "Times New
]),

// ████████████████████████████████████████████████████████████████████
// PAGE 3 – ACKNOWLEDGEMENT
// ████████████████████████████████████████████████████████████████████
section([
  heading1("ACKNOWLEDGEMENT"),
  ...sp(1),
```

```
      body("The satisfaction and euphoria that accompany the successful completion of any task would be incompl

  ),
      ...sp(1),
      body("My sincere thanks to the highly esteemed institution BANGALORE COLLEGE OF ENGINEERING AND TECHNOLOG
      ...sp(1),
      body("I express my sincere gratitude to Dr. GOPALAIAH, Principal, BCET, Bangalore, for providing the requ
      ...sp(1),
      body("I am extremely thankful to Mrs. KAVITHA R, HOD, Department of Computer Science and Engineering, BCE
      ...sp(1),
      body("I am grateful to Dr. RAMESH KUMAR B, who helped me to complete this Internship Program successfully
      ...sp(1),
      body("Finally, I am grateful to my parents and friends for the invaluable support, guidance, and encourag
      ...sp(3),
      new Paragraph({ alignment: AlignmentType.RIGHT, spacing: { before: 60, after: 40 },
        children: [new TextRun({ text: "NAME: SACHIN", font: "Times New Roman", size: 22 })] }),
      new Paragraph({ alignment: AlignmentType.RIGHT, spacing: { before: 40, after: 40 },
        children: [new TextRun({ text: "USN: 1BC22CS042", font: "Times New Roman", size: 22 })] }),
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // PAGE 4 – COMPANY CERTIFICATE (placeholder)
    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    section([
      new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 40, after: 20 },
        children: [new TextRun({ text: "i", font: "Times New Roman", size: 22, bold: true })] }),
      heading1("COMPANY CERTIFICATE"),
      ...sp(1),
      new Paragraph({ spacing: { before: 40, after: 40 },
        children: [new TextRun({ text: "➢  BESANT TECHNOLOGIES:", font: "Times New Roman", bold: true, size: 2
      ...sp(1),
      new Table({
        width: { size: 9000, type: WidthType.DXA },
        columnWidths: [9000],
        rows: [new TableRow({ children: [new TableCell({
          borders: THIN_BORDERS,
          width: { size: 9000, type: WidthType.DXA },
          margins: { top: 200, bottom: 200, left: 400, right: 400 },
          children: [
            new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 60, after: 60 },
              children: [new TextRun({ text: "INTERNSHIP COMPLETION CERTIFICATE", font: "Times New Roman", bold
            ...sp(1),
            new Paragraph({ alignment: AlignmentType.JUSTIFIED, spacing: { before: 60, after: 60, line: 360, li
              children: [
                new TextRun({ text: "This is to certify that ", font: "Times New Roman", size: 22 }),
                new TextRun({ text: "Sachin", font: "Times New Roman", bold: true, size: 22 }),
                new TextRun({ text: " from ", font: "Times New Roman", size: 22 }),
                new TextRun({ text: "Bangalore College Of Engineering And Technology", font: "Times New Roman",
                new TextRun({ text: ", has successfully completed an internship at Besant Technologies from ",
                new TextRun({ text: "02-02-2026 to 20-05-2026", font: "Times New Roman", bold: true, size: 22 
                new TextRun({ text: " in the role of ", font: "Times New Roman", size: 22 }),
                new TextRun({ text: "Data Science and Machine Learning", font: "Times New Roman", bold: true, s
                new TextRun({ text: " Intern.", font: "Times New Roman", size: 22 }),
              ]
            }),
            ...sp(1),
            body("During the internship, the individual's performance

  was observed to be sincere, dedicated, and enthusiastic towards learning and executing assigned responsibilitie
            ...sp(1),
            body("We wish them all the best in their future endeavors."),
            ...sp(1),
            body("This certificate is issued as a formal recognition of the successful completion of the intern
            ...sp(2),
            new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 60, after: 40 },
              children: [new TextRun({ text: "Branch Manager", font: "Times New Roman", bold: true, size: 22 })
            new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 20, after: 20 },
              children: [new TextRun({ text: "Besant Technologies, Electronic City", font: "Times New Roman", s
            new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 10, after: 10 },
              children: [new TextRun({ text: "Ganga Enclave, No. 7, 3rd Floor, Neeladri Rd,", font: "Times New
            new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 10, after: 10 },
              children: [new TextRun({ text: "Electronic City Phase I, Bengaluru, Karnataka 560100", font: "Tin
          ]
        })]})],
      }),
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // PAGE 5 – ABSTRACT
```

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  // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  section([
    new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 20, after: 10 },
      children: [new TextRun({ text: "ii", font: "Times New Roman", size: 22, bold: true })] }),
    heading1("ABSTRACT"),
    ...sp(1),
    body("Data Science and Machine Learning represent one of the most impactful fields in modern technology,
    ...sp(1),
    body("The internship was structured around building a Sales Intelligence Dashboard using the US Supersto
    ...sp(1),
    body("During the internship, the project involved data acquisition and cleaning, engineering features fo
    ...sp(1),
    body("Additionally, the project integrated the Google Gemini API to auto-generate executive summaries fr
    ...sp(1),
    body("This internship significantly enhanced technical and analytical skills, providing hands-on experie
  ]),

  // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  // PAGE 6 – TABLE OF CONTENTS
  // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  section([
    new Paragraph({ alignment: AlignmentType.LEFT, spacing: { before: 20, after:

10 },
      children: [new TextRun({ text: "iii", font: "Times New Roman", size: 22, bold: true })] }),
    heading1("CONTENTS"),
    ...sp(1),
    new Table({
      width: { size: 9360, type: WidthType.DXA },
      columnWidths: [1000, 6960, 1400],
      rows: [
        // header
        new TableRow({ children: [
          new TableCell({ borders: THIN_BORDERS, width: { size: 1000, type: WidthType.DXA }, margins: { top:
            children: [new Paragraph({ children: [new TextRun({ text: "TITLE", font: "Times New Roman", bold
          new TableCell({ borders: THIN_BORDERS, width: { size: 6960, type: WidthType.DXA }, margins: { top:
            children: [new Paragraph({ children: [new TextRun({ text: "", font: "Times New Roman", bold: tru
          new TableCell({ borders: THIN_BORDERS, width: { size: 1400, type: WidthType.DXA }, margins: { top:
            children: [new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: "PAGE
        ]}),
        tocRow("", "ACKNOWLEDGEMENT", "i"),
        tocRow("", "COMPANY CERTIFICATE", "ii"),
        tocRow("", "ABSTRACT", "iii"),
        tocRow("", "CHAPTERS", "PAGE NO"),
        tocRow("1.", "COMPANY PROFILE", ""),
        tocRow("", "1.1  About VTU Collaboration", "1"),
        tocRow("", "1.2  About Besant Technologies", "1–2"),
        tocRow("2.", "DATA SCIENCE AND MACHINE LEARNING", ""),
        tocRow("", "2.1  Introduction to Data Science", "3"),
        tocRow("", "2.2  Role of AI in Data Science", "3–4"),
        tocRow("", "2.3  Tools and Technologies Used", "5"),
        tocRow("3.", "INTRODUCTION", ""),
        tocRow("", "3.1  Problem Statement", "6"),
        tocRow("", "3.2  Objective of the Project", "7–8"),
        tocRow("", "3.3  Methodology", "9"),
        tocRow("4.", "IMPLEMENTATION AND RESULTS", ""),
        tocRow("", "4.1  Implementation Overview", "14–15"),
        tocRow("", "4.2  Results", "16–18"),
        tocRow("5.", "ADVANTAGES, DISADVANTAGES AND APPLICATIONS", ""),
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        tocRow("", "5.2  Disadvantages", "19"),
        tocRow("", "5.3  Applications", "20"),
        tocRow("6.", "CONCLUSION AND FUTURE ENHANCEMENTS", ""),
        tocRow("", "6.1  Conclusion", "21"),
        tocRow("", "6.2  Future Enhancements", "21–22"),
        tocRow("7.", "REFERENCES", "23"),
      ]
    }),
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  // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
  // PAGE 7 – LIST OF FIGURES
  // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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            figRow("Fig 4.2", "Sales vs Profitability Analysis by Category", "15"),
            figRow("Fig 4.3", "Top and Bottom 10 Products by Profit", "15"),
            figRow("Fig 4.4", "Profit Margin by Sub-Category", "16"),
            figRow("Fig 4.5", "Regional Performance Heatmap", "16"),
            figRow("Fig 4.6", "ML Model Forecasting Output (Gradient Boosting)", "17"),
            figRow("Fig 4.7", "Discount vs Profit Correlation", "17"),
            figRow("Fig 4.8", "Sales Intelligence Dashboard – Slides Output", "18"),
          ]
        }),
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      // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
      // CHAPTER 1 – COMPANY PROFILE
      // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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        heading2("COMPANY PROFILE"),
        heading3("1.1 About VTU Collaboration:"),
        body("Visvesvaraya Technological University (VTU) actively collaborates with various industry partners, i
        ...sp(1),
        body("Students benefit from these collaborations through hands-on experience with real-time projects, exp
        ...sp(1),
        heading3("1.2 About Besant Technologies"),
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          children: [new TextRun({ text: "➢  Besant Technologies", font: "Times New Roman", bold: true, size: 22
        body("Besant Technologies is a leading software training and placement institute in India that provides p
        ...sp(1),
        body("Besant Technologies has established itself as a significant player in the IT education and training
        ...sp(1),
        heading3("Key Offerings/Collaborations:"),
        bullet("VTU Collaboration: Besant Technologies is a prominent partner for Visvesvaraya Technological Univ

U), offering comprehensive internship training programs (often referred to as \"Besant internships\" or \"Skil
        bullet("Curriculum: Their training covers in-demand skills such as Data Science (Python, Pandas, NumPy, S
        bullet("Industry Integration: Training programs are designed based on the latest technologies and market
        bullet("Certifications: They aim to provide students with the skills and certifications necessary for suc
      ]),

      // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
      // CHAPTER 2 – DATA SCIENCE AND MACHINE LEARNING
      // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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        heading2("DATA SCIENCE AND MACHINE LEARNING"),
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        body("Data Science with Python is the process of collecting, organizing, analyzing, and interpreting larg
        ...sp(1),
        body("Python is the most widely used language in data science because of its simplicity, power, and rich
        ...sp(1),
        heading3("2.2 Role of AI in Data Science"),
        body("Artificial Intelligence (AI) is rapidly transforming the field of Data Science — not by replacing d
        ...sp(1),
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          children: [new TextRun({ text: "The key roles of AI in Data Science:", font: "Times New Roman", bold: t
        numbered("Automating Feature Engineering: AI can automatically detect relevant features, transform variab
        numbered("Predictive Modelling: AI-powered algorithms like Gradient Boosting, Neural Networks, and Random
        numbered("Natural Language Generation: Generative AI models (such as Gemini and GPT) can automatically dr
        numbered("Anomaly Detection: AI can continuously monitor data streams to flag outliers or unusual pattern
        numbered("Prescriptive Analytics: While traditional analytics answers \"what happened,\" AI-powered presc
        ...sp(1),
        heading3("Main Steps in Data Science with Python:"),
```

```
      numbered("Data Collection: Acquiring data from source

s such as databases, APIs, CSV files, or web scraping. The data may be structured or unstructured."),
      numbered("Data Cleaning: Correcting errors, handling missing values, removing duplicates, and standardizi
      numbered("Feature Engineering: Creating new variables from raw data, including lag features, rolling aver
      numbered("Exploratory Data Analysis (EDA): Applying visualizations and statistics to understand patterns,
      numbered("Model Training and Evaluation: Training multiple machine learning algorithms, comparing them us
      numbered("Deployment and Reporting: Serializing trained models, generating automated reports, and present
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // CHAPTER 2 continued – Tools
    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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      subbullet("To collect, clean, and transform raw data into analysis-ready datasets"),
      subbullet("To identify patterns, seasonal trends, and business insights through EDA"),
      subbullet("To build and evaluate predictive machine learning models"),
      subbullet("To generate automated executive summaries using Generative AI"),
      subbullet("To visualize data through professional charts, dashboards, and slide decks"),
      subbullet("To provide strategic recommendations that improve business performance"),
      ...sp(1),
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      new Paragraph({ spacing: { before: 40, after: 40 },
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      bullet("Pandas – Data manipulation and cleaning; reading CSV files and building analytical DataFrames."),
      bullet("NumPy – Numerical computing; array operations and mathematical transformations."),
      bullet("Matplotlib & Seaborn – Data visualization; creating 8 business-driven charts for EDA."),
      bullet("Scikit-learn – Machine learning model training, evaluation, and serialization using model.pkl."),
      bullet("Plotly – Interactive charting for the demo dashboard."),
      ...sp(1),
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      bullet("Google Gemini API – Generative AI for automated executive summary and slide narrative generation.
      bullet("Jupyter Notebook – Interactive development environment for sprint-based notebooks."),
      bullet("Matplotlib PDF Backend – For compiling presentation slides into slides.pdf."),
      bullet("Pickle (model.pkl) – For serializing and reusing the trained Gradient Boosting forecaster."),
      ...sp(1),
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      bullet("US Superstore Dataset – A real-world retail dataset from Plotly's public repository containing 9,
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // CHAPTER 3 – INTRODUCT

ION
    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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      body("The Sales Intelligence Dashboard is an end-to-end data science and AI system designed to transform
      ...sp(1),
      body("In traditional business reporting, analysts spend significant time manually inspecting spreadsheets
      ...sp(1),
      body("The project was developed across four structured sprints: Sprint 1 focused on Exploratory Data Anal
      ...sp(1),
      heading3("3.2 Problem Statement"),
      body("In competitive retail environments, organizations generate thousands of transactions monthly but la
      ...sp(1),
      body("Key challenges include:"),
      bullet("Revenue Forecasting Gap: Without predictive models, businesses cannot anticipate upcoming seasona
      bullet("Profitability Blind Spots: High-revenue categories like Furniture appear successful but frequent
      bullet("Discount Mismanagement: Sales teams apply discounts intuitively, unaware that discounts above 20%
      bullet("Regional Imbalance: Certain region-category combinations (e.g., Furniture in the East) are chroni
      bullet("Manual Reporting Bottleneck: Producing executive summaries from structured data requires signific
      ...sp(1),
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      numbered("Time-Series Data Leakage Prevention: Splitting chronological data for model training requires c
      numbered("Model Selection Complexity: Comparing three ML algorithms (Linear Regression, Ran
```

```
dom Forest, Gradient Boosting) on a small 48-month dataset requires robust cross-validation strategy."),
      numbered("Generative AI API Reliability: The Gemini API integration requires exponential backoff and retr
      numbered("Iterative Forecasting: Projecting 3 months beyond training data requires a sliding-window loop
    ]),

    // ████████████████████████████████████████████████████████████████████
    // CHAPTER 3 continued – Objectives + Methodology
    // ████████████████████████████████████████████████████████████████████
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      subbullet("Sales Revenue Forecasting using Machine Learning (Gradient Boosting Regressor)"),
      subbullet("Generative AI Executive Reporting via Google Gemini API"),
      subbullet("Eliminate Manual Report Generation Bottleneck"),
      subbullet("Deliver Strategic Business Recommendations through Data-Driven Analysis"),
      subbullet("Produce Reusable ML Pipeline via model.pkl Serialization"),
      ...sp(1),
      heading3("3.3 Methodology"),
      numbered("Requirement Analysis: Identified the need to analyze US Superstore retail data across 4 years.
      numbered("System Design: The project was divided into four sprint modules — EDA (Sprint 1), Model Traini
      numbered("Data Collection and Cleaning: The raw superstore.csv dataset (9,994 rows, 21 columns) was down
      numbered("Exploratory Data Analysis (Sprint 1): Eight business-driven charts were generated and saved as
      numbered("Feature Engineering for Forecasting: Monthly revenue was aggregated and the following features
      numbered("Model Training and Comparison (Sprint 2): Three regression models were trained on the engineere
      numbered("Iterative Revenue Forecasting: A sliding-window forecast loop was programmed to project revenue
      numbered("Generative AI Integration (Sprint 3): The Google Gemini API was integrated using a Chain-of-Tho
      numbered("Presentation Generation: The generate_slides.py script used Matplotlib's PDF backend (PdfPages)

art images with AI-generated bullet insights into a nine-slide slides.pdf executive presentation."),
      numbered("Demo Pipeline (Sprint 4): All four sprints were unified into a single 04_demo.ipynb notebook de
      numbered("Testing and Validation: Each notebook and script was tested against the full dataset. Model out
    ]),

    // ████████████████████████████████████████████████████████████████████
    // PLATFORM FLOW PAGE
    // ████████████████████████████████████████████████████████████████████
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            "Data Cleaning & Feature Engineering (clean_data.py)",
            "↓",
            "Exploratory Data Analysis → 8 Business Charts (01_eda_sales.ipynb)",
            "↓",
            "Model Training: Linear Regression / Random Forest / Gradient Boosting",
            "↓",
            "Best Model Selected: Gradient Boosting → Serialized as model.pkl",
            "↓",
            "Iterative Revenue Forecasting (3-Month Horizon)",
            "↓",
            "Google Gemini API → Auto-Generated Executive Summaries",
            "↓",
            "Slide Deck Compilation → slides.pdf (9 Slides)",
            "↓",
            "Unified Demo Pipeline (04_demo.ipynb)",
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        ]
      }),
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    // ████████████████████████████████████████████████████████████████████
    // CHAPTER 4 – IMPLEMENTATION AND RESULTS
```

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    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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      heading2("IMPLEMENTATION AND RESULTS"),
      heading3("4.1 Implementation Overview"),
      bullet("Data Acquisition and Cleaning: The Superstore CSV was downloaded from Plotly's public repository.
      bullet("Sprint 1 – Exploratory Data Analysis: Eight business visualizations were created and saved to pre
      bullet("Feature Engineering for Time-Series: Monthly revenue was grouped and six predictive features were
      bullet("Sprint 2 – Machine Learning Model Comparison: Three Scikit-learn regressors (Linear Regression, R

ological 80/20 splitting. Models were evaluated on RMSE and MAE metrics."),
      bullet("Model Serialization: The Gradient Boosting Regressor — identified as the best model — was pickled
      bullet("Iterative Forecasting Loop: A sliding-window loop was programmed to feed model predictions back a
      bullet("Sprint 3 – Generative AI Integration: The Google Gemini API was integrated with a Chain-of-Though
      bullet("Presentation Generation: generate_slides.py compiled the 8 chart images and AI-generated bullet p
      bullet("Sprint 4 – Unified Demo Pipeline: The 04_demo.ipynb notebook orchestrated the entire pipeline end
      bullet("Testing: All notebooks were tested against the full 9,994-row dataset. Forecasts were validated a
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // RESULTS
    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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            ...sp(1),
            body("Year-over-Year Revenue Growth: $484K (2014) → $733K (2017) — consistent upward trend confirm
            body("Seasonal Peaks: Strong Q4 spikes (November–December) observed every year, driven by holiday r
            body("Q1 Recurring Drop: January–March shows a systematic revenue dip each year, indicating a seaso
          ]
        })]})],
      }),
      ...sp(1),
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          margins: { top: 120, bottom: 120, left: 200, right: 200 },
          shading: { fill: "F8FAFC", type: ShadingType.CLEAR },
          children: [
            new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: "SPRINT 1: Profitab
            ...sp(1),
            body("Technology vs

 Furniture: Technology orders generate up to $8K single-transaction profit. Furniture shows high-value orders t
            body("Top Product: Canon imageCLASS MF7460 copier is the highest earning product at over $25K total
            body("Loss Leader: Cubify CubeX 3D Printer contributes over $8.8K in cumulative losses — the bigges
            body("Tables Sub-Category: Profit margin of -11.7% across all Table orders. Bookcases (-2.7%) and S
          ]
        })]})],
      }),
      new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 40, after: 80 },
        children: [new TextRun({ text: "Fig 4.2 & 4.3  Sales vs Profit Scatter | Top/Bottom Products [chart2, c
    ]),

    section([
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            new Paragraph({ alignment: AlignmentType.CENTER, children: [new TextRun({ text: "SPRINT 2: Machine
            ...sp(1),
            body("Best Model: Gradient Boosting Regressor — achieved lowest RMSE and MAE among the three candic
            body("Features Used: lag_1, lag_2, rolling_3, month_sin, month_cos, Orders count."),
            body("Data Split: Chronological 80/20 split (shuffle=False) to prevent temporal data leakage."),
            body("3-Month Forecast: Iterative sliding-window loop projects revenue for the 3 months beyond tra
            body("Model Artifact: Saved as model.pkl — fully reusable pipeline for future monthly data updates
          ]
        })]})],
      }),
      new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 40, after: 80 },
        children: [new TextRun({ text: "Fig 4.6  ML Model Forecasting Output (Gradient Boosting Regressor)", fo
      // Result 4 – Discount
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          shading: { fill: "F8FAFC", type: ShadingType.CLEAR },
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            ...sp(1),
            body("Discount-Profit Correlation: Negative correlation of -0.219 between discount rate and profit.
            body("Strategic Recommendation: Cap sales representative discounts at 15% maximum to protect transa
            body("Shipping Mode Insight: First Class shipping generates the highest average profitability per c
            body("Regional Heatmap: Technology is the top profit driver in East ($22K) and West ($20K). Furnitu
          ]

    })]})],
      }),
      new Paragraph({ alignment: AlignmentType.CENTER, spacing: { before: 40, after: 80 },
        children: [new TextRun({ text: "Fig 4.7 & 4.8  Discount vs Profit | Sales Intelligence Dashboard Slides
    ]),

    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
    // CHAPTER 5 – ADVANTAGES
    // ■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■■
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      heading3("5.1 Advantages"),
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  End-to-End Automa
      body("The entire pipeline from raw CSV ingestion to executive slide deck is fully automated. Running four
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Actionable Busine
      body("The dashboard delivers specific, quantified business insights — from exact loss-making products to
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Reusable ML Pipel
      body("The serialized model.pkl can be updated with new monthly data and immediately produce fresh forecas
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Generative AI Eff
      body("The Gemini API integration reduces executive summary writing time from hours to seconds, ensuring c
      ...sp(1),
      heading3("5.2 Disadvantages"),
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Limited Dataset S
      body("With only 48 monthly data points, the ML models have limited training data. Deep learning approache
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Gemini API Depend
      body("The AI summary generation requires an active internet connection and valid Gemini API key. The syst
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Static Dashboard"
      body("The output is a static PDF rather than an interactive real-time dashboard. Users cannot filter by c
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Single Dataset Sc
      body("The system is designed for the US Superstore schema. Adapting it to a different retail data schema
    ]),

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, font: "Times New Roman", bold: true, size: 22 })] }),
      body("Monthly revenue tracking, product profitability ranking, and discount policy optimization for retai
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Supply Chain and
      body("Seasonal revenue forecasts help procurement teams plan inventory levels before Q4 peaks and reduce
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Executive Busines
      body("AI-generated summaries and auto-compiled slide decks streamline the monthly business review process
```

```
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Sales Team Perfor
      body("Discount correlation analysis helps sales managers define evidence-based discount policies and meas
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "➢  Regional Strategy
      body("Regional heatmaps identify which product-region combinations are profitable, enabling targeted budg
    ]),

    // ██████████████████████████████████████████████████████████████████████
    // CHAPTER 6 – CONCLUSION
    // ██████████████████████████████████████████████████████████████████████
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      heading1("CHAPTER 6"),
      heading2("CONCLUSION AND FUTURE ENHANCEMENTS"),
      heading3("6.1 Conclusion:"),
      body("The Sales Intelligence Dashboard is a complete, production-grade data science solution that success
      ...sp(1),
      body("Key strategic findings include consistent year-over-year revenue growth with strong Q4 seasonality,
      ...sp(1),
      body("The Gradient Boosting Regressor emerged as the best forecasting model, and its serialization as mod
      ...sp(1),
      body("Overall, this internship provided deep practical exposure to the end-to-end data science workflow,
      ...sp(1),
      heading3("6.2 Future Enhancement:"),
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Real-Time Interac
      body("The static PDF presentation can be replaced with a live interactive Plotly Dash or Streamlit dashbo

er by region, category, date range, and customer segment in real time."),
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Advanced Forecast
      body("Future versions can incorporate Facebook Prophet or LSTM neural networks for more sophisticated sea
    ]),

    section([
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        children: [new TextRun({ text: "22", font: "Times New Roman", size: 22, bold: true })] }),
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Automated Anomaly
      body("An anomaly detection layer can be added to flag unusual transaction patterns in real time — such as
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Natural Language
      body("A conversational AI interface (powered by Gemini or GPT) can allow business users to query the dash
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Multi-Dataset Sup
      body("The pipeline can be generalized to support multiple retail dataset schemas through a configurable c
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Cloud Deployment
      body("The pipeline can be deployed on AWS or GCP with a monthly cron schedule, automatically ingesting fr
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Customer Segmenta
      body("K-Means or hierarchical clustering can be applied to customer transaction data to identify distinct
      new Paragraph({ spacing: { before: 40, after: 20 }, children: [new TextRun({ text: "❖  Advanced Report S
      body("A performance scoring system can be developed to auto-grade each product, region, and sales represe
    ]),

    // ██████████████████████████████████████████████████████████████████████
    // CHAPTER 7 – REFERENCES
    // ██████████████████████████████████████████████████████████████████████
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      heading2("REFERENCES"),
      ...sp(1),
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      numbered("Plotly US Superstore Dataset: https://raw.githubusercontent.com/plotly/datasets/master/supersto
      numbered("Scikit-learn Documentation – Gradient Boosting Regressor: https://scikit-learn.org/stable/modul
      numbered("Google Gemini API Documentation: https://ai.google.dev/gemini-api/docs"),

      numbered("Pandas Documentation – Time Series Analysis: https://pandas.pydata.org/docs/user_guide/timeseri
      numbered("Forecasting: Principles and Practice – Hyndman & Athanasopoulos (3rd ed., 2021)"),
      numbered("Python Machine Learning – Sebastian Raschka & Vahid Mirjalili (3rd ed., Packt Publishing)"),
      numbered("Matplotlib PdfPages Documentation: https://matplotlib.org/stable/api/backend_pdf_api.html"),
    ]),
  ]
});

Packer.toBuffer(doc).then(buffer => {
  fs.writeFileSync("/mnt/user-data/outputs/Sachin_Internship_Report.docx", buffer);
  console.log("Done! Sachin_Internship_Report.docx written.");
});
```